Bow Valley College

Course Outline

TECH9208

Networking Foundations
Winter 2023 - Current

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TECH9208 Networking Foundations

COURSE DESCRIPTION

The Networking Foundations course is designed for learners who want the knowledge and skills they need to begin a career in Information Technology Networking Infrastructure. Networking Foundations gives learners a broad, foundational understanding of networking, including wired and wireless networking, the protocols required to transmit data, and some common troubleshooting processes.

REQUISITES	None
EQUIVALENTS	None
CREDITS	0
HOURS	128
ELIGIBLE FOR PLAR	No

COURSE LEARNING OUTCOMES

Bow Valley College is committed to ensuring our graduates can demonstrate their abilities in key areas that will make them effective citizens and encourage their development as lifelong learners. In addition to the discipline-specific skills that learners acquire in their programs, the College has identified ten learning outcomes.

College-Wide Outcomes:

1. Communication
2. Thinking Skills
3. Numeracy and Financial Literacy
4. Working with Others
5. Digital Literacy
6. Positive Attitudes and Behaviours
7. Continuous Learning
8. Health and Wellness Awareness
9. Citizenship and Intercultural Competence
10. Environmental Sustainability

#	COURSE LEARNING OUTCOME(S)	COLLEGE WIDE OUTCOMES SUPPORTED
1	Explain the concept of network communication.	1, 2, 5
2	Build a functioning network.	2, 4, 5
3	Configure a Dynamic Host Configuration Protocol (DHCP) server.	2, 5
4	Configure an integrated wireless router and wireless client to connect securely to the Internet.	2, 5
5	Configure basic network security.	2, 5
6	Compare in-band and out-of-band management access.	2, 5
7	Build a simple computer network using both prohibitory protocol and open standards.	1, 2, 5
8	Troubleshoot basic network connectivity issues.	2, 5

COURSE MODULES AND SCHEDULE

**Course schedule subject to change, depending on delivery mode and term of study. For exact dates, please consult the Course Offering Information in Brightspace.*

WEEK/HOURS	MODULES
Week 1	Communications in a connected world, online connections, explore networks with Packet Tracer, build a simple Peer-to-Peer network, communication principles
Week 2	Network design and the access layer, routing between networks, the Internet Protocol (IP), dynamic addressing with DHCP, IPv4 and IPv6 address management
Week 3	Transport layer services, application layer services, build a home network, connect to the Internet, security considerations
Week 4	Configure network and device security, switches and routers, the internetworking operating systems (IOS) command line, build a small network, troubleshoot common network problems
Week 5	Final assessments

ASSESSMENT

COURSE LEARNING OUTCOME(S)	ASSESSMENT	WEIGHT
1, 2, 3, 4, 5, 6, 7, 8	Practical application	50%
1, 2, 3, 4, 5, 6, 7, 8	Final exam	50%

Important: For details on each assignment and exam, please see the Course Offering Information.

PERFORMANCE STANDARDS

A grade of P is required to pass this course. Learners may be required to pass this course in order to progress in the program or to meet specific program completion requirements.

To receive a course overall pass, the learner must pass all assessments.

Please consult with the program area or the Bow Valley College website for further details.

GRADING SCHEME

Symbol	**Description**	**Value in GPA**
P	Pass	N/A
F	Fail	N/A
BD	Basic/Developing (Foundational programming courses only)	N/A

REQUIRED LEARNING RESOURCES

Computer requirements:

Standards	**Hardware**	***Software***
Processor	i7	
RAM (memory)	16 GB RAM or greater	Windows 10 Pro/education 64-bit
Hard drive & storage	512 GB SSD or greater	(MacOS is not supported)
Video card	On-board integrated	
Screen size	15" or greater	

Additional learning resources may be found in the Course Offering Information or in Brightspace.

ADDITIONAL INFORMATION

Additional information may be found in the Course Offering Information or in Brightspace.

ACADEMIC ACCOMMODATIONS

Learners with a disability (learning, physical, and/or mental health) may qualify for academic and exam accommodations. For more information, or to apply for accommodations, learners should make an appointment with Accessibility Services in the Learner Success Services (LSS) Department. Accessibility Services can also assist learners who may be struggling with learning but do not have a formal diagnosis. To make an appointment visit LSS on the first floor of the south campus or call 403-410-1440. It is the learner's responsibility to contact Accessibility Services and request academic accommodations. For more information, please visit our website at http://www.bowvalleycollege.ca/accessibility.

INSTITUTIONAL POLICIES

Bow Valley College is committed to the highest standards of academic integrity and honesty. Learners are urged to become familiar with and uphold the following policies: Academic Integrity (500-1-7), Learner Code of Conduct, Procedures and Guidelines (500-1-1), Learner Appeals (500-1-12), Attendance (500-1-10), Grading (500-1-6), Academic Continuance and Graduation (500-1-5), and Electronic Communications (300-2-13). Audio or video recording of lectures, labs, seminars, or any other teaching and learning environment by learners is allowed only with consent of the instructor as part of an approved accommodation plan. Recorded material is to be used solely for personal study and is not being used or distributed without prior written consent from the instructor.

Turnitin:

Students may be required to submit their course work to Turnitin, a third-party service provider engaged by BVC. Turnitin identifies plagiarism by checking databases of electronic books and articles, archived webpages, and previously submitted student papers. Students acknowledge that any course work or essays submitted to Turnitin will be included as source documents in the Turnitin.com reference database, where it will be used solely to detect plagiarism. The terms that apply to a student's use of Turnitin are described on Turnitin.com.

Online Exam Proctoring:

Examinations for this course may require proctoring through an online proctoring service. Online

proctoring enables online exam taking within a controlled and monitored environment, thereby enhancing academic integrity. Online proctoring may occur through a variety of methods, including but not limited to:

a. live online proctoring where a remote invigilator authenticates identity and observes completion of an exam using specialized software and recordings;
b. automated proctoring where the exam session is recorded and AI (artificial intelligence) analyzed;
c. browser lockdown that limits access to other applications, websites, copying, printing, screen capture and other functions; or
d. a combination of both live/automated proctoring and browser lockdown.

Course instructors will review recordings, analyses, and data obtained through online proctoring for academic integrity infractions. It is the student's responsibility to meet the technical, software, location, and identity verification requirements necessary to enable online proctoring.

Further details of these policies are available in the Academic Calendar and on the Bow Valley College website, bowvalleycollege.ca.

Learners are encouraged to keep a copy of this course outline for future reference.

Collection of Personal Information:

This course, including your image and voice, may be recorded and made available to you and other students taking the course section. By attending the class(es) online or in person, you consent to the collection of your personal information. If you do not wish to be recorded, please contact your instructor before starting the course/class to discuss alternative arrangements.

You may use the recordings only for educational purposes and you must not copy, share, or use the recordings for any other purpose without the instructor's express permission.

Your personal information is collected in accordance with section 33(c) of the Freedom of Information and Protection of Privacy Act (Alberta) to deliver academic programming, support learner flexibility, promote universal design for learning principles, and for purposes consistent with the course activities and outcomes. If you have any questions about the collection, disclosure, use, or protection of this information, please contact the College's Access and Privacy Officer at foip@bowvalleycollege.ca.